



CSE 111: Structured Programming Language

Sidratul Tanzila Tasmi

Lecturer, Department of Computer Science and Engineering
United International University

Half Pyramid

```
*  
* *  
* * *  
* * * *  
* * * * *
```

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

```
A  
B B  
C C C  
D D D D  
E E E E E
```



Half Pyramid



```
*  
* *  
* * *  
* * * *  
* * * * *
```

i	0	1	2	3	4
0					
1					
2					
3					
4					

Half Pyramid



```
*
**
***
****
*****
```

i	0	1	2	3	4
0	*				
1					
2					
3					
4					

Half Pyramid

```
*  
* *  
* * *  
* * * *  
* * * * *
```

i	0	1	2	3	4
0	*				
1	*	*			
2					
3					
4					

Half Pyramid

```
*  
**  
***  
****  
*****
```

i	0	1	2	3	4
0	*				
1	*	*			
2	*	*	*		
3					
4					

Half Pyramid

```
*  
* *  
* * *  
* * * *  
* * * * *
```

i	0	1	2	3	4
0	*				
1	*	*			
2	*	*	*		
3	*	*	*	*	
4					

Half Pyramid

```
*  
* *  
* * *  
* * * *  
* * * * *
```

i	0	1	2	3	4
0	*				
1	*	*			
2	*	*	*		
3	*	*	*	*	
4	*	*	*	*	*

Half Pyramid



1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

i	0	1	2	3	4
0	1				
1	1	2			
2	1	2	3		
3	1	2	3	4	
4	1	2	3	4	5

Half Pyramid

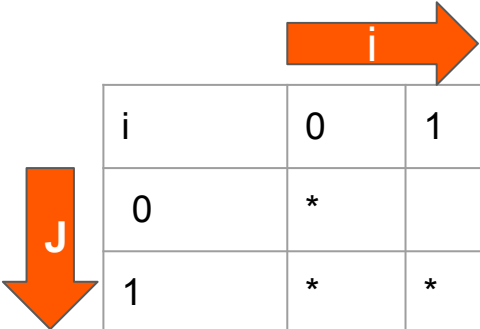


A
BB
CCC
DDDD
EEEE

i	0	1	2	3	4
0	A				
1	B	B			
2	C	C	C		
3	D	D	D	D	
4	E	E	E	E	E



Code

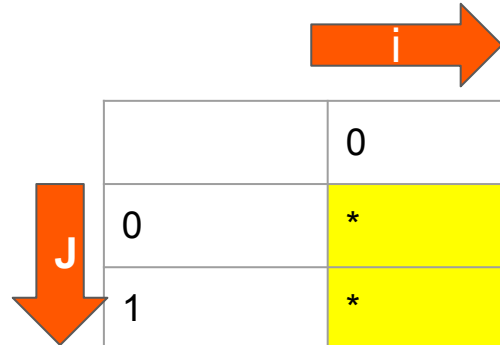


i	0	1	2	3	4
0	*				
1	*	*			
2	*	*	*		
3	*	*	*	*	
4	*	*	*	*	*

```
#include <stdio.h>
int main() {
    int i, j, rows;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; ++i) {
        for (j = 1; j <= i; ++j) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

Nested Loop Example

input	output
2	** **
4	**** **** **** ****



	0	1	2	3
0	*	*	*	*
1	*	*	*	*
2	*	*	*	*
3	*	*	*	*

Nested Loop Example

input	output
2	** **
4	**** **** **** ****

```
int main() {  
    int n;  
    scanf("%d",&n);  
    for(int i=0;i<n;i++){  
        for(int j=0;j<n;j++) {  
            printf("*");  
        }  
        printf("\n");  
    }  
}
```

Nested Loop Example

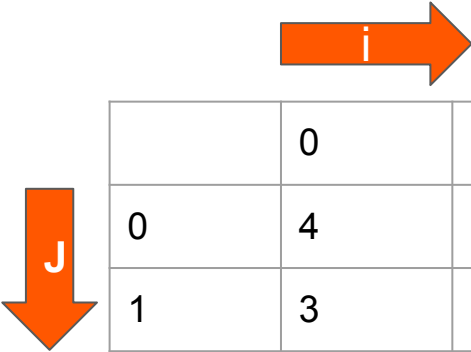
input	output
2	21 1
4	4321 321 21 1



	0	1	2	3
0	4	3	2	1
1	3	2	1	
2	2	1		
3	1			

Nested Loop Example

```
int main() {  
    int n;  
    scanf("%d",&n);  
    for(int i=n;i>=1;i--){  
        for(int j=i;j>=1;j--) {  
            printf("%d",j);  
        }  
        printf("\n");  
    }  
}
```



	0	1	2	3
0	4	3	2	1
1	3	2	1	
2	2	1		
3	1			

Nested Loop Example

input	output
2	12 23
4	1234 2345 3456 4567

```
int main() {  
    int n;  
    scanf("%d",&n);  
    for(int i=1; i<=n ;i++){  
        for (int j=0;j<4;j++){  
            printf("%d",i+j);  
        }  
        printf("\n");  
    }  
}
```

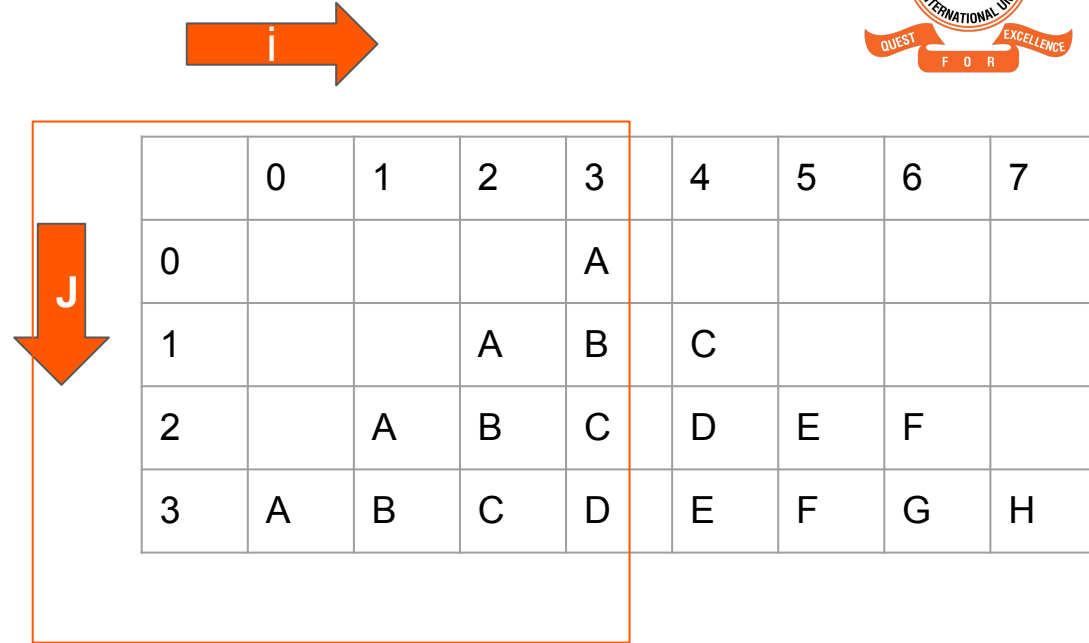
Nested Loop Example

input	output
3	<pre> A A B C A B C D E </pre>
4	<pre> A A B C A B C D E F A B C D E F G I </pre>

i	space	Pattern
1	$6 = 2 * 3$	$2*(4-1)$
2	$4 = 2 * 2$	$2*(4-2)$
3	$2 = 2 * 1$	$2*(4-3)$
4	$0 = 2 * 0$	$2*(4-4)$
		$2*(n-i)$

Nested Loop Example

input	output
3	<pre> A A B C A B C D E </pre>
4	<pre> A A B C A B C D E F A B C D E F G H </pre>



	0	1	2	3	4	5	6	7
0				A				
1			A	B	C			
2		A	B	C	D	E	F	
3	A	B	C	D	E	F	G	H

Nested Loop Example

For n=4

i	space	Pattern
1	6 = 2 * 3	$2*(4-1)$
2	4 = 2 * 2	$2*(4-2)$
3	2 = 2 * 1	$2*(4-3)$
4	0 = 2 * 0	$2*(4-4)$
		$2*(n-i)$



	0	1	2	3	4	5	6	7
0				A				
1			A	B	C			
2		A	B	C	D	E	F	
3	A	B	C	D	E	F	G	H

Nested Loop Example

input	output
3	<pre> A A B C A B C D E </pre>
4	<pre> A A B C A B C D E F A B C D E F G I </pre>

```

int main() {
    int n;
    char c;
    scanf("%d",&n);
    for(int i=1; i<=n ;i++){
        //printing leading whitespaces
        for(int s=1 ; s<= 2*(n-i); s++) printf(" ");
        //printing character
        c='A';
        for(int j=0 ; j< 2*i-1 ;j++) printf("%c ",c+j);
        printf("\n");
    }
}

```

Nested Loop Example



For $n=4$ and upper portion

input	output
3	<pre> * * * * * * * * * </pre>
4	<pre> * * * * * * * * * * * * * * * * </pre>

i	space	Pattern	Star
1	3	4-1	1
2	2	4-2	2
3	1	4-3	3
4	0	4-4	4
		$n-i$	i

Nested Loop Example



input	output
3	<pre> * * * * * * * * *</pre>
4	<pre> * * * * * * * * * * * * * * * *</pre>

For $n=4$ and lower portion

i	space	stars
1	1	3
2	2	2
3	3	1
	i	$n-i$

Nested Loop Example

input	output
3	<pre> * * * * * * * * *</pre>
4	<pre> * * * * * * * * * * * * * * * *</pre>

```
// for upper protion
for(int i=1; i<=n ;i++){
    //printing leading whitespaces
    for(int s=1 ; s <= (n-i); s++) printf(" ");
    //printing star
    for(int j=0 ; j<i ;j++) printf("* ");
    printf("\n");
}
```

```
// for lower protion
for(int i=1; i<=n-1 ;i++){
    //printing leading whitespaces
    for(int s=1 ; s <= i ; s++) printf(" ");
    //printing star
    for(int j=0 ; j< n-i ;j++) printf("* ");
    printf("\n");
}
```



Thank you!